

Computational Social Science Analysis

Course Catalogue Number

75250084FY

Credits

6 EC

Entry requirements

The course is open to students of the Research Master Social Sciences. PhD students from the Amsterdam Institute for Social Science Research (AISSR) can also apply for this course.

This is an advanced course which will involve programming in Python. While knowledge of programming is not a formal prerequisite to enter the course, the course will cover substantial ground, and you are therefore **strongly recommended** to prepare by for instance following a (free) online course. See for instance <https://www.codecademy.com/learn/python>.

Instruction language

English

Time Period(s)

2025-2026, semester 2, block 5

Times / Location

Lectures: Mondays 11:00-13:00.

Workshops: Wednesdays 15:00-17:00.

Collaborative work sessions: Fridays 13:00-17:00 (Room: JK3.02) [optional]

See for locations <https://rooster.uva.nl>.

Video recording

The lectures will not be recorded.

Lecturer(s)

Dr. Petter Törnberg p.tornberg@uva.nl – course coordinator and lecturer.

Michelle Schimmel m.schimmel@uva.nl – TA, lecturer and student support.

Learning Outcomes

After this course, you will be able to:

1. Articulate the foundational principles and significance of Computational Social Science (CSS) within the broader context of social science disciplines;
2. Demonstrate ability to use the Python programming language for basic computational tasks, including writing simple scripts and functions;
3. Extract clean data from real-world digital systems for computational analysis, using methods such as web scraping, APIs, and data cleaning;
4. Employ key analytical techniques such as Natural Language Processing (NLP), machine learning, and Social Network Analysis (SNA) effectively as part of a social scientific research project;
5. Evaluate the ethical considerations and challenges of computational research, focusing on data collection, analysis, and interpretation;

6. Plan a research project proposal that integrates well-formulated research questions, appropriate data sources, and outlines appropriate methods of analysis;
7. Critically evaluate the use of Large Language Models as tools for measurement, classification, and simulation in social science research;
8. Write a coherent and clear research paper presenting the project findings that adheres to academic standards of writing and citation.

These learning outcomes relate to the following exit qualifications of the RMSS:

1. The graduate has acquired knowledge and understanding:
 - a) Of the theories and relevant research methods and techniques within the candidate's chosen methodological specialization;
2. The acquired cognitive and/or professional abilities enable the graduate to apply:
 - a) Theoretical, epistemological and methodological knowledge and insight to the formulation and resolution of scientifically and socially relevant research problems in a disciplinary, multidisciplinary and/or interdisciplinary context;
 - c) In-depth knowledge of and experience with advanced research methods and techniques relevant to a specific research problem;
 - g) Skills to think and work in a scholarly fashion at a level of achievement in accordance with an internationally recognized research MSc standard;
4. The graduate should be able to use the following communicative skills:
 - b) Independently present social science research findings both orally and in writing in English to the required scholarly standard, making use of theoretical insights relevant to the particular research problem under study;
5. The graduate has acquired the following learning skills:
 - b) A critical attitude and the analytical and research skills needed to qualify for a PhD study after graduation or to function on a professional level in research and policy departments of governmental and non-governmental organizations.

Course Content

Computational Social Science (CSS) is a rapidly evolving field that combines computational methods, digital data, and social science theory to study social phenomena. Increasingly, this field is being reshaped by advances in artificial intelligence—particularly Large Language Models (LLMs)—which are transforming how social data is generated, interpreted, and analyzed. This course introduces students to CSS with a strong emphasis on how AI is changing both the objects of study and the methods used to study them. At the same time, AI is an emerging object of study in its own right, raising new questions about communication, knowledge production, labor, and power.

The RMSS course *Computational Social Science Analysis* covers a range of methods, including natural language processing, social network analysis, machine learning, social simulation, and the use of LLMs as tools for measurement, classification, and research design. Students will learn how to work with large-scale digital data from sources such as social media, online platforms, and digital archives, while also critically engaging with the opportunities and limitations introduced by AI-driven methods. Particular attention is given to how LLMs can be used in research workflows, as well as the challenges they raise for validity, bias, and interpretation.

By combining computational techniques with substantive social science questions, the course equips students to analyze complex social dynamics in a data-rich and AI-mediated environment. This approach prepares students for academic research as well as roles in industry and government, where understanding and critically deploying AI tools is becoming essential.

The course is strongly practice-oriented and offers a structured introduction to conducting and writing a computational social science research paper. Students work in teams of two and follow a process that mirrors academic research:

1. You will develop a research proposal, defining your question and research design.
 2. You will carry out an original research project and write it up as an academic paper.
 3. You will peer-review a paper by fellow students, and your own paper will receive two peer reviews.
 4. You will revise your paper based on feedback and submit a final version for grading.
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Teaching methods/learning formats

This course is project-based, and your grading will primarily be based on your independent research project carried out with a peer student. Your initial research paper proposal will also be graded. In the final week, you will peer-review another student project, and your review will also be part of your final grade. Each week consists of one lecture, and one lab session. During the lab sessions, you will go through code together with the instructor, and work on practical examples. Each week also has a longer work session, where you can work together to help each other, and receive support from the course TA.

Office hours

The lecturer is available for questions during breaks and after the lectures and workshops. Office hours are Mondays 13-15. Petter Törnberg's office is Room L6.38, LAB42, Science Park. Please schedule an appointment by email prior.

Teamwork

The first task is for students to find a collaborator and decide on a topic. To do so, think of the area and method that you would like to employ, and find someone with similar interests. Feel free to make use of the Canvas forum to post and find someone with similar interests and complementary method skills. **Note:** It is *your* responsibility to find a collaboration partner!

There will be a regular place to work together with your classmates **Friday between 13:00 and 17:00 in JK3.02**, during the entire course. You are strongly encouraged to make use of this space, and to support your peers throughout the course. Michelle Schimmel will be present to support you with your work and project.

Exceptions to the duo teamwork rule can be made in individual cases. If there is an uneven number of students or a student disenrolls the course, it may also be necessary for students to work individually on their projects, or in a group of three.

Student Feedback & Adjustments of the Course

Based on student feedback, the course has been modified so that projects are carried out in teams of two students, to encourage collaboration. The course content has also been adapted to better fit the skills of the students.

Manner & Form of Assessment and Assessment Criteria (Participation/Attendance/Effort Requirements)

The assessment consists of the following elements (please look closely at the [instructions and assessment details](#) at the end of this document):

- 20% Peer-review (individual)
- 80% Research paper (duo)

The grade furthermore involves a mandatory pass/fail Research Proposal hand-in (see below).

The passing grade is a 5.5. The [Rules and Guidelines for the Master's programme](#) apply. Re-sits and re-writes are only possible for students who have taken part in the original examination or have submitted a paper at the original deadline. In case of extenuating circumstances, the student can request the Examinations Board for an exception. In event of a fail, the maximum grade in the resit is 7.

Research paper proposal

The first task is for students to find a collaborator and decide on a topic.

For the initial project proposal task, student teams are expected to draft a concise **one-page document** (A4, 11pt) outlining their proposed research project within the realm of Computational Social Science.

The proposal should be a tentative plan for the story of the final paper. It should include an introduction, research question, preliminary hypothesis or set of hypotheses, an outline of the computational techniques and data sources, and references to preliminary literature. The grading will be **pass/fail**. If the student fails, they will have the chance to revise their document based on feedback from the instructor.

Peer review

After having finished the draft of their paper, every student will carry out peer reviews of two other student research papers. Your peer review should be **around 1 page in length** (A4, Times New Roman, 11pt). The grading will be 0-10 points.

Final research paper

Your project will result in a brief research paper, written in the style of a scholarly journal article. This paper should be **4000 ± 200 words in length** (including abstract and figure text but excluding references) and clearly present the research gap, research question, the methodologies used, ethical considerations, and the findings you have drawn, and discussions of their implications. The grading will be 0-10 points.

You will hand in your assignments through Canvas.

Overview assignment deadlines

	Deadline date
Research paper proposal	2026-04-13 (before class begins: 11:00)
Research paper draft (to reviewers)	2026-05-20 (before 12:00)
Peer-reviews	2026-05-22 (before end of workday: 17:00)
Final research paper submission	2026-05-27 (before end of workday; 17:00)
[Paper Resit/Repair]	2026-06-29 (before end of workday; 17:00)

Attendance requirements

The lectures and workshops are necessary for learning within this course. The RMSS has a strict attendance requirement. This means that students can miss 2 out of 12 meetings. Only in case of extenuating circumstances students can miss more classes. Please show respect to your peers and your lecturer by being on time.

Students who are unable to attend a meeting need to inform the lecturer beforehand. Students who do not meet this requirement and are absent without a valid reason can be expelled from further participation.

If you are absent for reasons outside your control, please inform the lecturer.

The Friday group work is not mandatory, but attending is highly recommended as they provide the opportunity to practice together and receive support when questions arise.

Inspection/Perusal of exams/assignments, feedback

Grades will be released within 15 working days after the deadlines. The research paper proposal will receive individual written feedback. Your research paper draft will receive peer feedback from two other students. Feedback on the final research paper and on your peer feedback will be provided in written form along with the final grade.

Rules regarding Fraud and Plagiarism

Plagiarism of any kind will not be tolerated. Plagiarism not only involves the direct and intentional copying of existing texts without attribution, but is often the result of sloppy citation and referencing, or via self-plagiarism. Self-plagiarism is when students submit the same work for the fulfilment of multiple degree requirements. Electronic detection software will be used to detect plagiarism. In submitting a text, the student implicitly consents to the text being entered into the database of the detection programme concerned. More information about plagiarism and its consequences can be found in the UvA regulatory guidelines:

<https://student.uva.nl/en/topics/plagiarism-and-fraud>

Use of AI-based writing tools

The recent rise of AI-based writing and programming tools (e.g., ChatGPT, Claude, Copilot) has made their use a standard professional skill. You are therefore encouraged to use such tools in this course, both for coding and for writing. At the same time, all submitted work must reflect your own understanding and intellectual ownership.

If you use AI tools, you must include a brief note in the Acknowledgements section (before the reference list) specifying which tools you used and how you used them.

Core principle: you must understand your work.

If you submit code, analysis, or text, you are responsible for understanding what it does, being able to explain it clearly, and ensuring it accurately represents your work. Using AI to support your learning is encouraged; submitting work you do not understand is not.

Acceptable use includes debugging, explaining concepts, improving structure, and generating or refining small parts of code or text that you review and adapt. You should treat AI as an assistant, not as a substitute for doing the work.

Unacceptable use includes submitting AI-generated code or text without understanding it, generating entire assignments with AI, misrepresenting what your code does, including fabricated or incorrect references, or presenting outputs you cannot reproduce or explain.

All computational assignments must be submitted as a Jupyter Notebook with clear structure, commented code, and reproducible outputs. The notebook should make your reasoning transparent.

If there are concerns about your understanding, you may be asked to attend a short 1-on-1 discussion where you walk through your code and explain your approach. Failure to demonstrate sufficient understanding will affect your grade.

Fabricated references, misleading or dishonest representation of methods or results, or submitting work you cannot explain will be treated as academic misconduct. In research, this would be grounds for desk rejection; in this course, it may result in a failing grade.

In short: use AI to learn faster, not to bypass learning.

Please read the most updated UvA guidelines and make sure that you are compliant:

<https://student.uva.nl/en/topics/ai-tools-and-your-studies>

Literature/materials

We will not follow a particular coursebook, but each lecture will come with recommended reading. The recommended readings for the lectures are optional but are important starting points for your project, which will require you to go deeper into specific methods. The readings for each lecture will be available on Canvas.

As part of working on your individual project, it is your responsibility to identify and do the additional readings that you need.

Date Final Grade

The final grades will be available to students 15 working days after the submission of the final research paper at the latest. The final grades also need to be available at least 10 working days before a resit.

Schedule

Week	Dates	Lecture	Workshop	Project work	Holidays / Notes
1	30 Mar – 3 Apr	What is Computational Social Science? + How to do a research project	Introduction to Python	Programming homework and proposal writing	Good Friday 3 Apr (no teaching Fri)
2	6 Apr – 10 Apr	Acquiring Big Data: Scraping and APIs + Ethics & Law	Web Scraping and APIs in practice	Programming and proposal writing continued	Easter Monday 6 Apr (no teaching Mon)
3	13 Apr – 17 Apr	Natural Language Processing for the Social Sciences	Natural Language Processing in practice	Individual project work	Proposal due Mon 13 Apr, 11:00

4	20 Apr – 24 Apr	Social Network Analysis	Social Network Analysis in practice	Individual project work	
—	27 Apr – 1 May	<i>No teaching or exams</i>			King's Day Mon 27 Apr
5	4 May – 8 May	Machine Learning in the social sciences	Machine Learning in practice	Individual project work	Collective day off Mon 4 May; Liberation Day Tue 5 May
6	11 May – 15 May	Agent-Based Modeling and Online Experiments	Agent-Based Modeling in practice + How to write a research paper	Individual project work	Ascension Day Thu 14 May; Collective day off Fri 15 May
—	18 May – 22 May	<i>No lecture</i>		Draft & peer review	Draft due Wed 20 May 12:00; Peer-reviews due Fri 22 May 17:00
—	25 May – 29 May	<i>Assessment / submission</i>			Whitsun Mon 25 May; Final paper due Wed 27 May 17:00
—	29 Jun				Resit/Repair deadline Mon 29 Jun 17:00

WEEK 1

Lecture: What is Computational Social Science? + How to do a research project

This lecture will provide a feeling for what Computational Social Science is and situate it within other social scientific disciplines. We will discuss “Big Data” and how it is different from traditional qualitative and quantitative social science data. You will also get an overview of the methods that exists within CSS, what they can do, and get some first inspiration on what your course project can be.

We will also talk about your research project, and how to write a paper in just 6 weeks.

Workshop: Introduction to Python programming

This workshop provides a first introduction to Python programming. During the first two weeks, you will be working on a Python notebook for basic Python programming.

Project work: Programming homework and proposal writing

During the first two weeks, you will be working intensely on your research proposal, while also working through a Python programming course. You will receive individual support from the instructor in relation to the workshop.

Recommended readings:

1. Lazer, D. M., Pentland, A., Watts, D. J., Aral, S., Athey, S., Contractor, N., ... & Wagner, C. (2020). Computational social science: Obstacles and opportunities. *Science*, 369(6507), 1060-1062. <https://www.science.org/doi/full/10.1126/science.aaz8170>
2. Törnberg, P., & Uitermark, J. (2021). For a heterodox computational social science. *Big Data & Society*, 8(2). <https://doi.org/10.1177/20539517211047725>

3. Caleb Ziems, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang. 2024. Can Large Language Models Transform Computational Social Science?. *Computational Linguistics*, 50(1):237-291. <https://aclanthology.org/2024.cl-1.8/>

WEEK 2

Lecture: Acquiring Big Data: Scraping and APIs + Ethics & Law

This first half of this lecture gives an introduction to the process of acquiring digital data for social science research from various online sources. We focus on existing data sources, explore APIs, and webscraping. We will also discuss data cleaning and preparation.

The second half of the lecture focuses on ethical considerations and legal frameworks that govern the collection, analysis, and dissemination of digital data. The lecture focuses on practical skills, such as applying for an IRB review, discuss ethics in a research paper, and complying with GDPR.

Workshop: Web Scraping and APIs in practice

In this workshop, we will learn to scrape data. We will first go through coding examples for web scraping. You will be provided with code that you can adapt to your individual project.

Project work: Programming and proposal writing *continued*

Recommended readings:

1. Salganik, M. J. (2018). *Bit by bit: Social research in the digital age*. Princeton University Press. [Chapter 6: Ethics.] <https://www.bitbybitbook.com/en/1st-ed/ethics/>
2. El-Sayed, S. & Paspalj, F. (2024). "No recognised ethical standards, no broad consent: navigating the quandary in computational social science research." *Big Data & Society*. <https://doi.org/10.1177/17470161241247686>
3. New York Times. "The Secretive Company That Might End Privacy as We Know It". 2020. <https://www.nytimes.com/2020/01/18/technology/clearview-privacy-facial-recognition.html>

WEEK 3

Deadline 1: Hand in research project plan.

Lecture: Natural Language Processing

The lecture will give an overview of Natural Language Processing: from text preprocessing, vectorization, to a wide array of NLP techniques, ranging from sentiment analysis to named entity recognition, topic modeling, and machine learning approaches. The focus is on how the methods can be practically employed and be used to answer social scientific research questions.

Workshop: Natural Language Processing in practice

We will go through a series of important NLP methods and use them to analyze an example dataset.

Project work: Students will work on their individual projects.

Recommended readings:

1. Gentzkow, M., Kelly, B. & Taddy, M. (2019). "Text as Data." *Journal of Economic Literature*, 57(3), 535–574. <https://doi.org/10.1257/jel.20181020>
2. Hobson Lane, Cole Howard, Hannes Hapke (2024) *Natural Language Processing in Action: Understanding, analyzing, and generating text with Python*. Manning. Chapter 1-4.

3. Daniel Jurafsky and James H. Martin. 2024. *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*.
https://web.stanford.edu/~jurafsky/slp3/ed3bookfeb3_2024.pdf
4. Petter Törnberg. 2023. How to use LLMs for Text Analysis. *Arxiv*.
<https://arxiv.org/abs/2307.13106>
5. Petter Törnberg. 2024. Best Practices for Text Annotation with Large Language Models. *Arxiv* <https://arxiv.org/abs/2402.05129>

WEEK 4

Lecture: Social Network Analysis

This lecture focuses on Social Network Analysis: a methodological approach that examines the patterns of relationships and interactions among social entities. This session is designed to introduce students to the fundamental concepts and tools of SNA, including network theory, metrics for analyzing network structures (such as centrality, density, and clustering coefficients), and the visualization of complex social networks.

Workshop: Social Network Analysis in practice

In this workshop, we will learn how to create a social network from real-world data, and analyze it using Gephi and networkx.

Project work: Students will continue work on their individual projects.

Recommended readings:

1. John Scott. 2011. Social Network Analysis: Development, Advances, and Prospects. *Social Network Analysis and Mining*. <https://link.springer.com/article/10.1007/s13278-010-0012-6>
2. Mustafa Emirbayer. 1997. Manifesto for a Relational Sociology. *American Journal of Sociology*.
<https://doi.org/10.1086/231209>
3. Johan Ugander, Brian Karrer, Lars Backstrom, Cameron Marlow. 2011. “The Anatomy of the Facebook Social Graph”. arXiv. <https://arxiv.org/abs/1111.4503>
4. Soroush Vosoughi, Deb Roy, Sinan Aral. 2018. “The spread of true and false news online”. *Science*. <http://doi.org/10.1126/science.aap9559>
5. John Scott. 2012. *What is Social Network Analysis?*
<https://library.oapen.org/bitstream/handle/20.500.12657/58730/1/9781849668200.pdf>
[For those interested in going deeper.]

WEEK 5

Lecture: Machine Learning in the social sciences

This lecture focuses on machine learning and the many ways in which it can be employed in the social sciences. The discussion begins with a foundational overview of machine learning — from supervised and unsupervised learning to state-of-the-art deep learning — and proceeds to illuminate how these algorithms can be employed to identify patterns, predict outcomes, and provide insights in vast datasets that were previously unmanageable for human researchers.

Workshop: Machine Learning in practice

In this workshop, we will go through code that shows how machine learning can be used in practice in the social sciences.

Project work: Students will continue work on their individual projects.

Recommended readings:

1. Molina, M., & Garip, F. (2019). Machine learning for sociology. *Annual Review of Sociology*, 45, 27-45. <https://www.annualreviews.org/doi/abs/10.1146/annurev-soc-073117-041106>
2. Grimmer, J., Roberts, M. E., & Stewart, B. M. (2021). Machine learning for social science: An agnostic approach. *Annual Review of Political Science*, 24, 395-419. <https://www.annualreviews.org/doi/abs/10.1146/annurev-polisci-053119-015921>
3. James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An introduction to statistical learning: With applications in python*. Springer Nature.
4. Athey, S. & Imbens, G. (2019). "Machine Learning Methods That Economists Should Know About." *Annual Review of Economics*, 11, 685–725. <https://doi.org/10.1146/annurev-economics-080217-053433>

WEEK 6

Lecture: Agent-Based Modeling + Online Experiments

The first half of this lecture will focus on Agent-Based Modeling: a way of simulating social systems to identify the link between individual behavior and macro-level outcomes.

The second half the lecture will focus on online experiments, in which we build platforms to examine how specific interventions shape behavior.

Workshop: Agent-Based Modeling in practice, and How to Write a Paper

This session will focus on implementing a simple Agent-Based model – the Schelling segregation model – and analyze its dynamics and implications.

The workshop will also include a lecture with advice on how to write and tell an exciting story through a research paper.

In the final 15 minutes, we will also fill in the course evaluation form using the Evalytics.

Project work: Students will continue work on their individual projects.

Recommended readings:

1. Schelling, T. C. (1971). Dynamic models of segregation. *Journal of mathematical sociology*, 1(2), 143-186. <https://www.tandfonline.com/doi/abs/10.1080/0022250X.1971.9989794>
2. Epstein, J. M. (2008). "Why Model?" *Journal of Artificial Societies and Social Simulation*, 11(4), 12. <https://www.jasss.org/11/4/12.html>
3. Petter Törnberg, Diliara Valeeva, Justus Uitermark, Christopher Bail. (2023) Simulating Social Media Using Large Language Models to Evaluate Alternative News Feed Algorithm. *Arxiv*. <https://arxiv.org/abs/2310.05984>
4. Salganik, M. J. (2018). *Bit by bit: Social research in the digital age*. Princeton University Press. [Chapter 4: Running experiments.] <https://www.bitbybitbook.com/en/1st-ed/running-experiments/>
5. Salganik, M. J., Dodds, P. S., & Watts, D. J. (2006). Experimental study of inequality and unpredictability in an artificial cultural market. *Science*, 311(5762), 854-856. DOI: <http://doi.org/10.1126/science.1121066>
6. Schimel, J. (2012). *Writing science: how to write papers that get cited and proposals that get funded*. OUP USA.

Instructions and assessment details

1. Instructions for Research Paper Proposal

Your first task is to write a one-page project proposal outlining your research idea. Think of it as a pitch for the story you want to tell in your final paper.

The primary goal is to make sure you have a feasible and interesting project idea, and to get early feedback from the lecturer on how to pursue it.

A key tip: Write the proposal as if you have already carried out the project. Imagine you are telling the story of your findings and framing them within existing scholarship. This forward-looking approach will help you identify gaps in your plan and clarify what you actually need to do.

Your proposal should include:

- A brief introduction to the topic, explaining why it matters and what it contributes to existing research.
- A clearly stated research question that is specific enough to be answerable within the course timeframe.
- A preliminary hypothesis or expected findings (if applicable).
- The data sources you plan to use and how you intend to access them.
- The computational methods you plan to apply and why they are appropriate for your question.
- A few key references situating your project in the literature.

Use the proposal to ask for help. If you are unsure about which method or data source to use, or how to frame your story, say so in the proposal. This is your opportunity to get tailored advice from the lecturer before you commit to a direction.

Format: Max 1 page (A4, 11pt). Submit as a team via Canvas.

Grading: Pass/fail. If you do not pass, you will have the opportunity to revise and resubmit based on written feedback from the lecturer. You will receive individual written feedback on your proposal in Week 3. You are also welcome to reach out to the lecturer for further advice at any point.

2. Instructions for Computational Social Science Research Paper

In this course, you are tasked with carrying out a research project using Computational Social Science (CSS) methods. The culmination of your project will be a short research paper, written in the style of a scholarly journal article. This paper should **be 4000 ± 200 words in length** (including abstract, but excluding reference list and the response to the peer-reviews; see below) and clearly present the research you conducted, the methodologies used, and the findings you have drawn. This project is an opportunity for you to delve into an area of interest within computational social science and demonstrate your ability to conduct and communicate research effectively. The paper represents 80% of the final course grade. The grade will be the same for both students. Other than the word length guidelines, write your paper aimed for the journal “Journal of Computational Social Science”:

Journal submission guidelines: <https://link.springer.com/journal/42001/submission-guidelines>

Journal aims and scope: <https://link.springer.com/journal/42001/aims-and-scope>

You are allowed to draw on any approach in your article: your paper does not need to be “typical” CSS paper but can be qualitative, quantitative, mixed-methods, or whatever you prefer. However, as outlined in the aims and scope, the journal only accepts papers that use some CSS-style methods or data, so there needs to be some central element of use of the methods that we have discussed during this course. For instance, you can carry out a sentiment analysis on YouTube comment data, and then either feed the result into a statistical model or use it for a qualitative-style interpretive analysis. Be creative! However, your approach needs to be clearly specified in your paper, and your paper needs to be written in an appropriate manner.

Tip: look at similar research papers published in academic journals and see how they are structured and how they make their argument. Be conscious of your audience and communicate your approach so as to situate your paper in a research field.

Sections of the Paper:

You do not need to use the headers below, but the paper should follow the following general structure.

1. **Title and Abstract**
 - **Title:** Concise and descriptive.
 - **Abstract:** 150-250 words summarizing the research question, methodology, results, and significance.
2. **Introduction**
 - Present the research question and its importance.
 - Briefly review relevant literature.
 - State the objectives of your research.
3. **Methodology**
 - Describe the CSS methods used, including data sources, analytical tools, and processes.
 - Justify your choice of methodology.
 - One or two paragraph discussing and motivating the **ethics** of your approach.
4. **Results**
 - Present your findings in a clear, logical order.
 - Use appropriate visuals (graphs, tables) where necessary.
 - Interpret the results in a concise manner.
5. **Discussion**
 - Interpret your findings in the context of the research question and literature review.
 - Discuss limitations and potential biases in your study.
 - Suggest areas for future research.
6. **Conclusion**
 - Summarize key findings and their implications.
 - Relate back to the objectives stated in the introduction.
7. **References**
 - Include all sources cited, following a standard academic format (e.g., APA, MLA).
8. **Appendices (optional)**
 - Include supplementary material that is relevant but not integral to the main text.
9. **Response to peer-reviews**
 - The final section should include a half-page section in which you respond to the comments from the peer-reviewers.

Response to Peer-Reviews

As part of your final paper submission, it's essential to include a detailed response to the peer-review comments you received on your initial draft. This response should not only acknowledge each comment but also explain how you addressed it in your final revision. For comments you chose not to incorporate, provide a brief explanation for your decision. Structure your response clearly by describing each reviewer's comment followed by your reply. This process demonstrates your engagement with the feedback and your commitment to improving your work based on peer insights. This thoughtful reflection and revision based on peer feedback are crucial elements of the scholarly writing process and contribute significantly to the quality of your final paper. It's crucial to respond to the reviewers in a respectful and considerate way that acknowledges their perspectives and the significant work that they invested in helping you improve your paper.

Length: The peer-review response should be between **500-600 words long**.

Detailed Instructions:

- **Research Topic:** Ensure that your topic is feasible given the time frame and resources available.
- **Data and Analysis:** Your project should demonstrate the ability to collect, analyze, and interpret data using CSS methods. Emphasize data-driven insights.
- **Writing Style:** Use clear, concise academic prose. Avoid jargon unless necessary and provide explanations for specialized terms.
- **Format:** Follow the guidelines of a specific academic journal in your field in terms of style, citation, and formatting.
- **Originality:** Your work must be original. Plagiarism will result in a failing grade and reporting to the Academic Integrity Committee.
- **Collaboration:** Collaboration and discussion with your peers is encouraged. The project is carried out in teams of two.

Note: Write as accessibly as possible. In the real world, papers are often reviewed by researchers with limited time and focus. While your experience as a student may have been that writing using complex language produces better grades, this strategy will rarely work in the real world, and it will not work in this course. Research papers must be written **as clearly and simple as possible**. If the reviewer or grader struggles to understand some aspect of it, your paper will receive a poor grade. When you peer-review in this course, make sure that you actually understand what the author is trying to get across.

Grading Criteria:

- **Clarity and coherence (20%):** The paper should be well-organized, with a logical flow of ideas.
- **Research and analysis (20%):** Depth of research, appropriateness of CSS methods, analysis of data, and ethical argumentation.
- **Interpretation of results (20%):** How well you interpret and discuss your findings.
- **Writing quality (15%):** Grammar, clarity, and adherence to journal style.
- **Overall impact (10%):** Originality, contribution to the field, and the practical significance of your findings.
- **Response to reviewers (15%):** Convincing response to reviewers and appropriate revisions of the paper to address issues that they raise.

Submission:

Draft:

You will first submit a draft of your paper for peer-review from fellow students. This draft should be in a readable and complete state to enable useful peer-review.

- **Format:** Submit via Canvas. The file format should be PDF or DOCX.
- **Late Submissions:** Late paper submission is not accepted, since it will not leave time for the peer review process. *Don't wait until last minute! Submit well on time!*

Final submission for grading:

- **Format:** Submit via Canvas. The file format should be PDF or DOCX.

When submitting your final version, you must also submit the code and data used in the project. This is standard when submitting research papers for publication. Please upload this on a code sharing page such as Github, and provide the URL in your paper.

3. Instructions and Advice for Student Peer-Reviewing

After having finished the draft of our paper, every student will carry out a peer review of one other student research paper. This is an individual task.

The peer review process is essential in academic research. It ensures the quality, validity, and significance of the research. As a peer reviewer, your role is to critically analyze and provide constructive feedback on the research paper, helping the author improve their work.

Your peer-review is anonymous.

Your peer reviews will be graded and will account for 20% of your total final grade, and is individually graded. Your peer review should be **around 1 page in length** (A4, Times New Roman, 11pt).

Your chief goal of this peer review is to provide the author with feedback on how they can best improve their paper. Your peer review will be graded on the extent to which it helps improve the paper, based on the criteria and advice outlined below. The grading will be 0-10 points.

Review Process

1. **Look at the paper grading criteria and journal guidelines:** Your role is to evaluate the paper on the basis of the course's grading criteria. It's therefore important that you look at the specific grading criteria and try to put yourself in the shoes of the professor. To do so, make sure to re-read the course grading criteria.
2. **Initial Reading:** Skim the paper to get an overall sense of its structure, argument, and main points.
3. **Detailed Reading:** Read the paper thoroughly. Take notes on major points, strengths, and areas needing improvement.

Evaluation Criteria

Look at the evaluation criteria and instructions for the research paper and use this as starting point for your peer review. Try to put yourself in the role of the grader and use this lens to provide constructive and kind feedback that will help the author get as high grade as possible on their paper. The ability of putting yourself in the role of an evaluator is a key skill in almost any job and will help you become better at your future academic or professional role.

1. **Content and Originality:**
 - Is the research topic relevant and original?
 - Does the paper contribute new knowledge or insights to the field?
2. **Structure and Organization:**
 - Is the paper well-organized with a clear introduction, methodology, results, discussion, and conclusion?
 - Are the arguments logically structured and easy to follow?
3. **Methodology:**
 - Are the methods used appropriate and well-explained?
 - Is the data analysis rigorous and valid?
4. **Ethics:**
 - Are sufficient ethical considerations made? Are the ethical dimensions of the data and methods employed sufficiently discussed?
5. **Results and Discussion:**
 - Are the results presented clearly and supported by data?
 - Does the discussion provide an accurate interpretation of the results?
6. **Style and Writing Quality:**

- Is the writing clear, concise, and free of grammatical errors?
 - Are technical terms and jargon used appropriately?
 - **Note:** If you struggle to understand anything in the paper, consider this an issue with the paper. As a reviewer, it is your task to point out if the paper is unclear or hard to read.
7. **References:**
- Are sources cited appropriately and relevant to the research?

Providing Feedback

Your task is to articulate criticism and feedback in a way that is helpful, constructive, and empathetic – a valuable skill both inside and outside academia. Constructive feedback helps authors understand the strengths and weaknesses of their work more clearly. It goes beyond merely pointing out flaws, but seeks to provide specific, actionable suggestions that can guide authors to improve their research.

1. **Be Constructive:** Offer both positive feedback and constructive criticism. Suggest concrete ways to improve the paper.
2. **Be Specific:** Provide specific examples from the paper when giving feedback. Avoid vague comments. Be specific about what can be improved and how. Example: Instead of saying "The methodology section is weak," suggest "The methodology section could be strengthened by providing more details on data collection procedures."
3. **Be Objective:** Focus on the content and quality of the paper, not on the author.
4. **Balance the Positive and the Constructive:** Ensure your feedback is balanced. Too much criticism can be disheartening, while only positive feedback may not be helpful for improvement. Aim for a constructive tone that motivates the author to make improvements.
5. **Focus on the Content and Structure:** Address key aspects such as clarity of argument, coherence of the narrative, adequacy of the evidence, logical flow, and relevance of the conclusions. Suggest reorganizing sections if the flow of the paper is disrupted.
6. **Comment on the Writing and Presentation:** Provide feedback on the clarity of the writing. Suggest where simplifying jargon, rephrasing sentences, or correcting grammatical errors would be beneficial. Encourage adherence to academic standards and formatting guidelines.
7. **Refer to the Guidelines and Criteria:** Refer to specific guidelines or assessment criteria for the paper in your feedback. Point out where and how the paper could better meet these criteria.
8. **Be open and respectful to diverse perspectives and ways of thinking:** Research takes many forms, and there are many right ways of doing things. Be careful to not confuse your preferred way with the right way.

Writing the Review

1. **Summary:** Start with a brief summary of the paper and your overall impression.
2. **Major Issues:** Highlight any major issues that need addressing.
3. **Minor Issues:** Point out minor issues such as typos, formatting errors, or unclear sentences.
4. **Final Thoughts:** Conclude with your overall assessment.

Post-Review

- **Submission:** Submit your review within the agreed timeline.
- **Follow-Up:** Be prepared to discuss your review with the “editor” or provide further clarification if needed.

While peer-reviews in the real world decide whether the paper will get published or not (e.g., “reject”, “major revisions”, “accept”), all papers in this course will naturally be “revise & resubmit”. You do not need to provide a grade on the paper or a recommendation for publication. Remember, your role as a peer reviewer is to help the author improve their work. Your thorough, fair, empathetic, and constructive feedback is invaluable in maintaining the integrity and quality of academic research. The author should feel empowered and supported to improve their paper.

Note: The presence of any unkind, harsh, disrespectful, or unconstructive language is the fastest way to a failing grade on this task.

Submission

- **Format:** Submit via Canvas. The file format should be PDF or DOCX.